THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

Assignment No.5

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1. Write a simple base class, then a derived class and use objects of both of them in the main function. It will be a simple illustration of inheritance.

```
#include <iostream>
using namespace std;


class Person {
public:
string name;
int age;

    void getPersonData() {
cout << "Enter name: ";
cin >> name;
cout << "Enter age: ";
cin >> age;
    }

    void displayPersonData() {
cout << "Name: " << name << endl;
cout << "Age: " << age << endl;
    }
};

class Student :
public Person {
public:
    int rollNo;

    void getStudentData() {
cout << "Enter roll number: ";
cin >> rollNo;
    }

    void displayStudentData() {
cout << "Roll Number: " << rollNo << endl;
    }
};
```

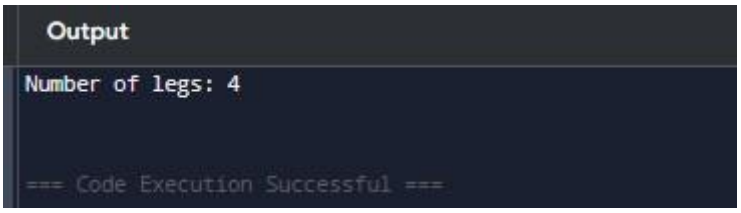
Output

```
Enter Person Details
Enter name: Ridhika
Enter age: 19

Person Details:
Name: Ridhika
Age: 19

------------------------
Enter Student Details
Enter name: Jaibhev
Enter age: 19
Enter roll number: 28

Student Details:
Name: Jaibhev
Age: 19
Roll Number: 28


=== Code Execution Successful ===
```

```
int main() {
Person p;
cout << "Enter Person Details\n";
p.getPersonData();
cout << "\nPerson Details:\n";
p.displayPersonData();

 cout << "\n------------------------\n";


Student s;
cout << "Enter Student Details\n";
 s.getPersonData();     // inherited function
 s.getStudentData();

 cout << "\nStudent Details:\n";
 s.displayPersonData(); // inherited function
 s.displayStudentData();
 return 0;
}
```

2. Practice protected access specifier in inheritance. In the base class declare a variable which is protected and access it in the derived class.

```
#include <iostream>
using namespace std;

class Animal {
protected:
    int legs;

public:
    void setLegs(int l) {
legs = l;
    }
};

class Dog : public Animal { public:
```

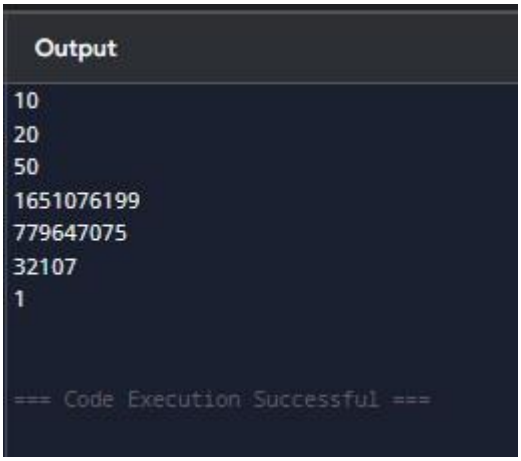


```
    void display() {         cout << "Number of legs:
" << legs << endl;   }
};

int main() {

    Dog d;
    d.setLegs(4);
    d.display();


    return 0;
}
```

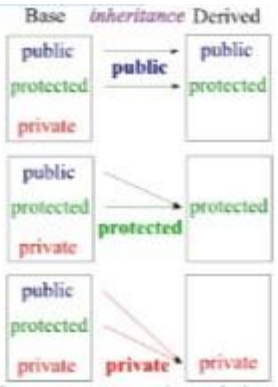
3. Most of the time we use public mode of inheritance, for example class Derived: public Base{}; Try protected and private access modifiers to understand the difference of various modes of inheritance.

```
#include <iostream>
using namespace std;

class Base {
public:
    int a;

protected:
    int b;

private:
int c;

public:
    void setValues() {
        a = 10;
b = 20;         c
= 30;
    }

    void showBase() {
```

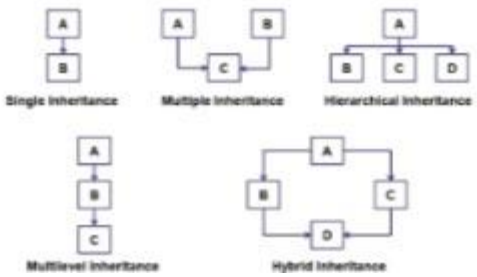
```
cout << "a = " << a << endl;
cout << "b = " << b << endl;
cout << "c = " << c << endl;
    }
};

class DerivedPublic : public Base { public:
    void display() {
cout << a << endl;
cout << b << endl;
    }
};

class DerivedProtected : protected Base { public:
    void display() {
cout << a << endl;
cout << b << endl;
    }
};
class DerivedPrivate : private Base { public:
    void display() {
cout << a << endl;
cout << b << endl;
    }
};
int main() {

    DerivedPublic obj1;
obj1.setValues();
obj1.display();
    obj1.a = 50;     cout <<
obj1.a << endl;

    DerivedProtected obj2;
obj2.display();

    DerivedPrivate obj3;
obj3.display();
    return 0;
}
```



4.



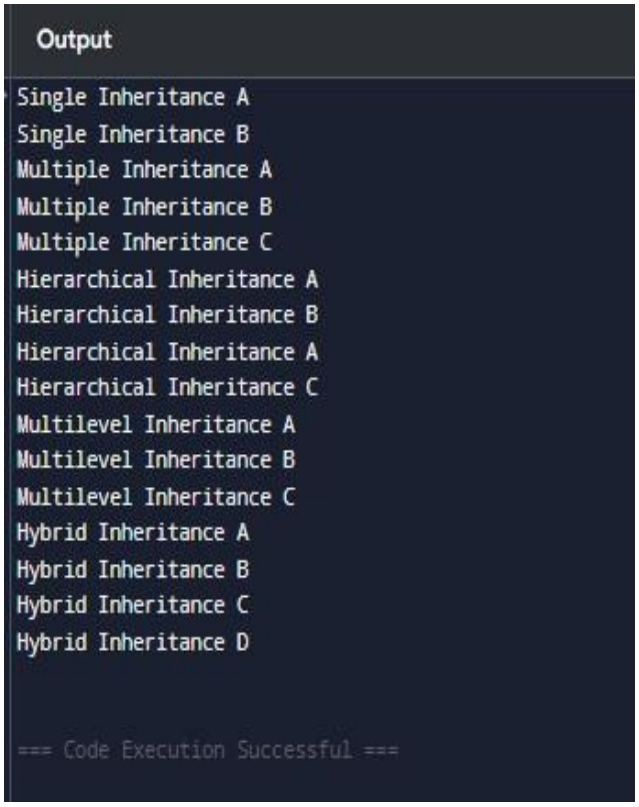
Various types of inheritances are as shown below. Write small C++ codes for each inheritance type.




```
#include <iostream>
using namespace std;

class A1 {
public:
        void showA1() {
        cout << "Single Inheritance A" << endl;
        }
};

class B1 :
public A1 {
public:
void showB1() {
cout << "Single Inheritance B" << endl;
}
};

class A2 {
public:
```

```
void showA2() {
cout << "Multiple Inheritance A" << endl;
}
};

class B2 {
public:
void showB2() {
cout <<
"Multiple
Inheritance B"
<< endl; }
};

class C2 : public A2, public B2 { public:     void showC2() {
cout << "Multiple Inheritance C" << endl; }
};

class A3 { public:     void showA3() { cout << "Hierarchical
Inheritance A" << endl; }
};

class B3 : public A3 { public:     void showB3() { cout <<
"Hierarchical Inheritance B" << endl; }
};

class C3 : public A3 { public:     void showC3() { cout <<
"Hierarchical Inheritance C" << endl; }
};

class A4 { public:     void showA4() { cout << "Multilevel
Inheritance A" << endl; }
};

class B4 : public A4 { public:     void showB4() { cout <<
"Multilevel Inheritance B" << endl; }
};

class C4 : public B4 { public:     void showC4() { cout <<
"Multilevel Inheritance C" << endl; }
};
```

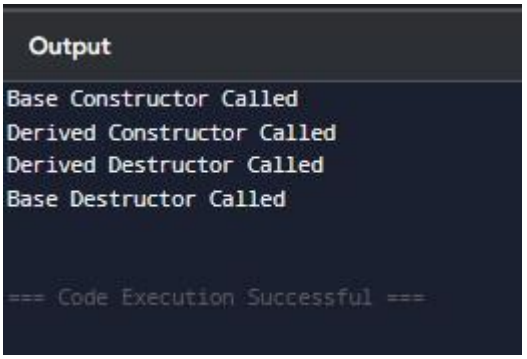
```
class A5 { public:
void showA5() {
cout << "Hybrid
Inheritance A"
<< endl; }
};

class B5 : virtual public A5 { public:     void showB5() { cout
<< "Hybrid Inheritance B" << endl; }
};

class C5 : virtual public A5 { public:     void showC5() { cout
<< "Hybrid Inheritance C" << endl; }
};

class D5 : public B5, public C5 { public:     void showD5() {
cout << "Hybrid Inheritance D" << endl; }
};

int main() {

   B1 obj1;
obj1.showA1();
obj1.showB1();

   C2 obj2;
obj2.showA2();
obj2.showB2();
obj2.showC2();

   B3 obj3;     C3
obj4;
obj3.showA3();
obj3.showB3();
obj4.showA3();
obj4.showC3();

   C4 obj5;
obj5.showA4();
obj5.showB4();
```



```
obj5.showC4();     D5
obj6;     obj6.showA5();
obj6.showB5();
obj6.showC5();
obj6.showD5();

    return 0;
}
```

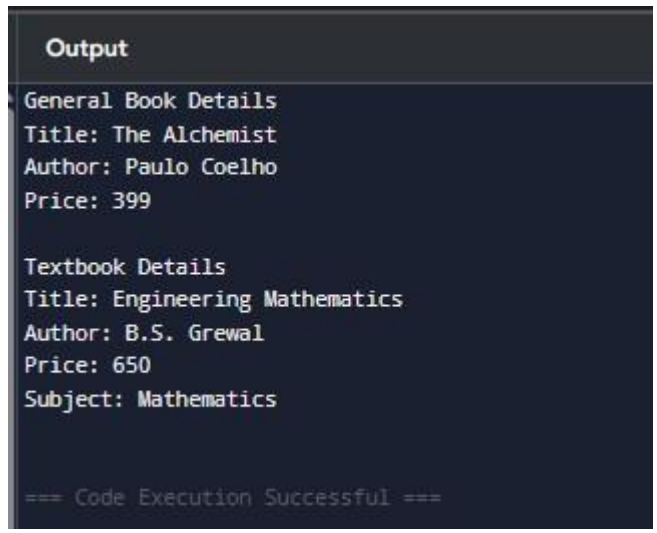
5. How can you use constructors and destructors in C++ inheritance? Write a program to illustrate.

```
#include <iostream>
using namespace std;

class Base { public:     Base() {         cout <<
"Base Constructor Called" << endl;
    }

    ~Base() {         cout << "Base Destructor
Called" << endl;
    }
};

class Derived : public Base { public:     Derived() {
cout << "Derived Constructor Called" << endl;
    }

    ~Derived() {         cout << "Derived Destructor
Called" << endl;
    }
};

int main() {

    Derived obj;

    return 0;
}
```



6. Implement a base class Book with attributes title, author, and price. Then, create a derived class Textbook that inherits from Book and adds a new attribute subject. Demonstrate how single inheritance is used to manage the data for general books and textbooks.

```
#include <iostream>
using namespace std;

class Book {
protected:
    string title;
string author;
float price;

public:
    void setBook(string t, string a, float p) {
title = t;          author = a;          price = p;
    }

    void displayBook() {          cout <<
"Title: " << title << endl;          cout <<
"Author: " << author << endl;          cout <<
"Price: " << price << endl;
    }
};

class Textbook : public Book { private:
    string subject;

public:
    void setTextbook(string t, string a, float p, string s) {
setBook(t, a, p);          subject = s;
    }

    void displayTextbook() {
        displayBook();          cout << "Subject:
" << subject << endl;
    }
};
```

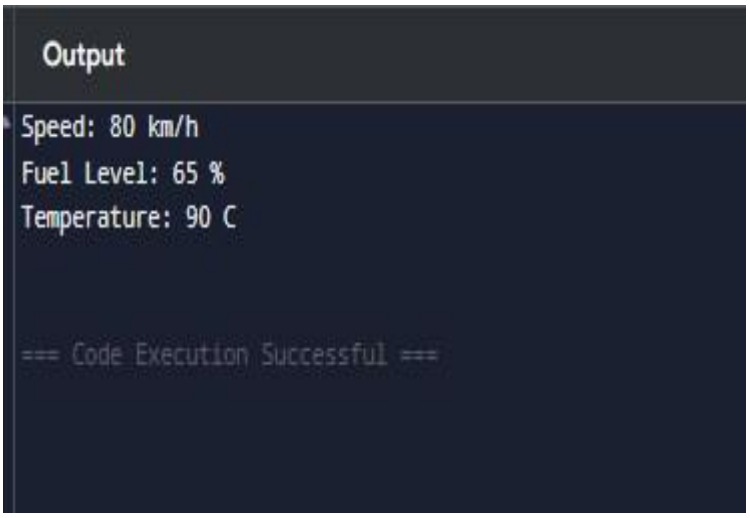


```
int main() {

    Book b;
    b.setBook("The Alchemist", "Paulo Coelho", 399);
cout << "General Book Details" << endl;
b.displayBook();

    cout << endl;

    Textbook tb;     tb.setTextbook("Engineering Mathematics", "B.S. Grewal", 650,
"Mathematics");     cout << "Textbook Details" << endl;     tb.displayTextbook();

    return 0;
}
```

7. A software company is creating a program to simulate a car's dashboard. The dashboard needs to display speed, fuel level, and temperature. The speed is controlled by a Speedometer class, the fuel level by a FuelGauge class, and the temperature by a Thermometer class. Implement the three classes:
Speedometer, FuelGauge, and Thermometer, each with relevant attributes and methods. Then, create a CarDashboard class that inherits from all three classes to display the combined information on the dashboard. Demonstrate how multiple inheritance is used to build this class.

```
#include <iostream>
using namespace std;
class Speedometer {
protected:
    float speed;
public:
    void setSpeed(float s) {
speed = s;
    }
    float getSpeed() {
return speed;
    }
};

class FuelGauge {
protected:
```

```
    float fuelLevel;
public:
    void setFuelLevel(float f) {
fuelLevel = f;
    }

    float getFuelLevel() {
return fuelLevel;
    }
};

class Thermometer {
protected:
    float temperature;
public:
    void setTemperature(float t) {
temperature = t;
    }

    float getTemperature() {
return temperature;
    }
};

class CarDashboard : public Speedometer, public FuelGauge, public Thermometer { public:
    void displayDashboard() {         cout << "Speed: " << speed
<< " km/h" << endl;         cout << "Fuel Level: " << fuelLevel
<< " %" << endl;         cout << "Temperature: " <<
temperature << " C" << endl;
    }
};

int main() {
    CarDashboard car;
    car.setSpeed(80);
car.setFuelLevel(65);
car.setTemperature(90);
    car.displayDashboard();
    return 0;
}
```



8. You are tasked with creating a system for a library that tracks different types of users. The system needs to handle general user information such as name, ID, and contact details. There are two specific types of users: Student and Teacher. Each type of user has additional attributes, such as grade level for students and department for teachers. Implement a base class LibraryUser with general attributes. Then, create two derived classes Student and Teacher that inherit from LibraryUser and add their own specific attributes. Demonstrate how hierarchical inheritance is applied in this scenario.

```
#include <iostream>
using namespace std;

class LibraryUser {
protected:     string
name;     int id;
string contact;

public:
   void setUser(string n, int i, string c) {
name = n;         id = i;         contact = c;
   }

   void displayUser() {         cout <<
"Name: " << name << endl;         cout <<
"ID: " << id << endl;
       cout << "Contact: " << contact << endl;
   }
};

class Student : public LibraryUser { private:
   string gradeLevel;

public:
   void setStudent(string n, int i, string c, string g) {
setUser(n, i, c);         gradeLevel = g;
   }

   void displayStudent() {         displayUser();
cout << "Grade Level: " << gradeLevel << endl;
   }
};
```

```
class Teacher : public LibraryUser { private:
    string department;

public:
    void setTeacher(string n, int i, string c, string d) {
setUser(n, i, c);         department = d;
    }

    void displayTeacher() {         displayUser();
cout << "Department: " << department << endl;
    }
};

int main() {

    Student s;
    s.setStudent("Ridhika",   280,   "7632490862",   "2nd   yr");
cout << "Student Details" << endl;     s.displayStudent();

    cout << endl;

    Teacher t;
    t.setTeacher("Mrs. Bansal", 301, "9123456780", "Electronics");
cout << "Teacher Details" << endl;     t.displayTeacher();

    return 0;
}
```

Output

```
Student Details
Name: Ridhika
ID: 280
Contact: 7632490862
Grade Level: 2nd yr

Teacher Details
Name: Mrs. Bansal
ID: 301
Contact: 9123456780
Department: Electronics


=== Code Execution Successful ===
```

9. A logistics company needs a software system to manage its vehicle fleet. All vehicles share common attributes like make, model, and year. Trucks have additional attributes like load_capacity. Furthermore, refrigerated trucks have a special attribute called temperature_control. Implement a base class Vehicle with common attributes. Then, create a derived class Truck that adds the load_capacity attribute. Finally, create another derived class RefrigeratedTruck that inherits from Truck and adds the temperature_control attribute. Demonstrate how multilevel inheritance works in this case.

```
#include <iostream> using
namespace std;

class Vehicle { protected:
   string make;
string model;     int
year;

public:
   void setVehicle(string mk, string md, int y) {
make = mk;         model = md;         year = y;
   }

   void displayVehicle() {         cout <<
"Make: " << make << endl;         cout <<
"Model: " << model << endl;         cout <<
"Year: " << year << endl;
   }
};

class Truck : public Vehicle { protected:
   float load_capacity;

public:
   void setTruck(string mk, string md, int y, float lc) {
setVehicle(mk, md, y);
      load_capacity = lc;
   }

   void displayTruck() {         displayVehicle();         cout << "Load
Capacity: " << load_capacity << " tons" << endl;
   }
};

class RefrigeratedTruck : public Truck { private:
   float temperature_control;

public:
   void setRefrigeratedTruck(string mk, string md, int y, float lc, float temp) {
setTruck(mk, md, y, lc);         temperature_control = temp;
   }
```

```
   void displayRefrigeratedTruck() {        displayTruck();        cout <<
"Temperature Control: " << temperature_control << " C" << endl;
   }
};

int main() {

   RefrigeratedTruck rt;     rt.setRefrigeratedTruck("Tata",
"Ultra", 2022, 12.5, -5);     rt.displayRefrigeratedTruck();

   return 0;
}
```

10. You are developing a software system for an academic institution. The institution has various roles like Person, Staff, and Student. A Person has general attributes like name and address. Staff members, who are a type of Person, have additional attributes like employee_id and department. Student, another type of Person, has attributes like student_id and grade. Some Staff members are also Students (e.g., teaching assistants) and need to inherit from both classes. Implement a base class Person with general attributes. Then, create derived classes Staff and Student that inherit from Person and add their specific attributes. Finally, create a TeachingAssistant class that inherits from both Staff and Student. Demonstrate how hybrid inheritance is applied and managed in this scenario.

```
#include <iostream>
using namespace std;

class Person { protected:
   string name;     string address;

public:
void setPerson(string n, string a) {
name = n;
address = a;
   }

   void displayPerson() {
cout << "Name: " << name << endl;
cout << "Address: " << address << endl;
   }
```

```
};

class Staff :
virtual public Person {
protected:
int employee_id;
string department;

public:
    void setStaff(int id, string dept) {
    employee_id = id;
     department = dept;
    }

void displayStaff() {
cout << "Employee ID: " << employee_id << endl;
cout << "Department: " << department << endl;
    }
};

class Student :virtual public Person {
protected:
int student_id;
string grade;

public:
void setStudent(int id, string g) {
student_id = id;
grade = g;
    }

    void displayStudent() {
cout << "Student ID: " << student_id << endl;
cout << "Grade: " << grade << endl;
    }
};

class TeachingAssistant : public Staff, public Student {
 public:
    void setTeachingAssistant(string n, string a, int eid, string dept, int sid, string g) {
setPerson(n, a);
```

```
        setStaff(eid, dept);
setStudent(sid, g);
    }

    void displayTeachingAssistant() {
displayPerson();
displayStaff();
displayStudent();
    }
};

int main() {

 TeachingAssistant ta;
ta.setTeachingAssistant("Ridhika", "Patiala", 280, "Electronics", 1024150, "A");
ta.displayTeachingAssistant();
    return 0;
}
```

Output

```
Name: Ridhika
Address: Patiala
Employee ID: 280
Department: Electronics
Student ID: 1024150
Grade: A


=== Code Execution Successful ===
```